The Anatomy of an Equation

Architectural role, epistemic status, and the classification of physical equations

Paper III of IV

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Abstract

A physical equation carries two independent pieces of information: where it sits within a theory's architecture, and what kind of claim it makes about the world. Paper I of this series [1] developed the architectural axis, classifying equations as structural, constitutive, dynamical or interactional within Tonti's grammar of configuration-kind and source-kind variables. The present essay develops the orthogonal axis: the epistemic status of an equation, whether it functions as a definition, an identity, an empirical law, a constitutive assumption, an approximation, an idealisation, a boundary condition, a conservation law, or a derived result. The same equation may shift position under a change of formulation, a change of theory, or a change of pedagogical context. Several case studies are examined in detail: $\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0$ as simultaneously an identity and a physical law; $\boldsymbol{F} = m\boldsymbol{a}$ as a fusion of definition, constitution and dynamics; the second law of thermodynamics as occupying different epistemic positions depending on the formulation adopted; and the equation of state $pV = nRT$ as a constitutive closure whose status as law, definition or idealisation shifts with the theoretical frame.

Contents

The epistemic axis

Paper I argued that a physical theory is a modular architecture in which equations occupy distinct functional positions: structure equations construct derived quantities, constitutive equations translate between descriptions, dynamical equations govern change, and interaction equations couple sectors [1]. The architecture was developed through five instantiated theories and was shown to explain the hinge phenomenon in physics pedagogy [4, 5].

That classification answers the question *where does this equation sit?* A second question is equally important: *What kind of claim is this equation making?* These two questions are orthogonal. An equation that occupies a structural position in the architecture may be a definition, a mathematical identity, or an empirical law. An equation that occupies a constitutive position may be an exact theoretical relation, a linear approximation, or a regime-dependent model assumption. The same mathematical statement can therefore be located at different positions in the matrix depending on the theoretical context in which it is read.

This second axis is where the present account extends beyond Tonti's classification. His grammar locates equations within a theory [2, 3]. The additional classification asks what authority each equation possesses, and how that authority changes under a change of formulation.

Epistemic status	What the equation does
Definition	Introduces a quantity by specifying how it is constructed from other quantities. Example: $\boldsymbol{v} = \mathrm{d}\boldsymbol{r}/\mathrm{d}t$.
Mathematical identity or representation	True by virtue of mathematical structure, independent of physical content. Example: $\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \boldsymbol{A}) = 0$.
Empirical law	An experimentally established regularity that could in principle have been otherwise. Example: $\boldsymbol{F} = q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B})$.
Constitutive law or model assumption	A relation encoding material, regime-dependent or theory-specific content. Example: $\boldsymbol{D} = \varepsilon \boldsymbol{E}$ for a linear isotropic dielectric.
Conservation or balance law	An equation expressing that some quantity is conserved, balanced or transported. Example: $\mathrm{d}\boldsymbol{p}/\mathrm{d}t = \boldsymbol{F}_{\mathrm{net}}$.
Approximation	A statement known to hold within a limited regime, understood as a controlled simplification of a more complete relation. Example: $\sin\theta \approx \theta$ for small angles.
Idealisation	A statement introducing a simplification that is not expected to hold exactly but that renders the model tractable. Example: treating a string as massless, or a surface as frictionless.
Boundary or initial condition	A specification of the state at a boundary or at an initial time that supplements the governing equations. Example: $\phi \to 0$ as $r \to \infty$.
Axiom or postulate	A statement adopted as a primitive constraint within a formulation, from which further results are developed. Example: an axiomatic statement of the second law used to define an entropy ordering.
Derived result	A consequence obtained by combining other equations, inheriting its authority from theirs. Example: $\nabla^2\phi = -\rho_q/\varepsilon$.

Table 1: The epistemic axis: what kind of claim an equation makes. The categories are not always exclusive.

The categories are proposed as a working taxonomy. They are not mutually exclusive in every case, and part of the interest lies precisely in equations that straddle or shift between categories.

The two-axis matrix

The result is a matrix rather than a single hierarchy. Each equation in a physical theory can, in principle, be assigned coordinates on both axes simultaneously. Table 2 gives the detailed classification; Figure 1 renders the same claim spatially, so that architectural movement and epistemic movement are visibly distinct.

Equation	Architectural role	Epistemic status
$\boldsymbol{v} = \mathrm{d}\boldsymbol{r}/\mathrm{d}t$	Structure	Definition of velocity from trajectory
$\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}$	Structure	Potential representation, locally available under the field constraint $\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0$
$\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0$	Structural constraint	Empirical field law within classical electromagnetism; a mathematical identity only after the vector-potential representation has been adopted
$\boldsymbol{p} = m\boldsymbol{v}$	Constitutive	Regime-dependent classical law, replaced by $\boldsymbol{p} = \gamma m_0 \boldsymbol{v}$ in relativistic mechanics
$\boldsymbol{D} = \varepsilon \boldsymbol{E}$	Constitutive	Material model, carrying linear and isotropic assumptions
$\mathrm{d}\boldsymbol{p}/\mathrm{d}t = \boldsymbol{F}$	Dynamical balance	Physical law relating force to momentum change
$\boldsymbol{F} = q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B})$	Interaction	Empirical coupling law
$\rho_q = q\delta(\boldsymbol{x} - \boldsymbol{r})$	Interaction / source map	Point-particle idealisation
$\nabla^2 \phi = -\rho_q/\varepsilon$	(none: composite)	Derived result, fusing structure, constitution and source in a homogeneous medium

Table 2: Nine equations classified on both axes simultaneously. The two columns are independent: equations sharing an architectural role may have different epistemic statuses, and vice versa.

architectural role	epistemic status: definition / representation	empirical / constitutive	balance / interaction	idealisation / derived result
structure	$\boldsymbol{v} = \mathrm{d}\boldsymbol{r}/\mathrm{d}t$ $\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}$	$\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0$ physical constraint		$\nabla^2 \phi = -\rho_q/\varepsilon$ composite result
constitution		$\boldsymbol{p} = m\boldsymbol{v}$ $\boldsymbol{D} = \varepsilon \boldsymbol{E}$		$pV = nRT$ ideal-gas model
dynamics			$\mathrm{d}\boldsymbol{p}/\mathrm{d}t = \boldsymbol{F}$	
interaction		$\boldsymbol{F} = q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B})$		$\rho_q = q\delta(\boldsymbol{x} - \boldsymbol{r})$ point-particle idealisation

Figure 1: The two classifications represented as genuinely independent axes. Placement is contextual rather than immutable: an equation may move horizontally when its epistemic status changes, and vertically when a reformulation changes its architectural role.

The distinction also connects to a persistent concern in physics education research. diSessa's account of *phenomenological primitives* describes intuitive knowledge as a collection of loosely connected explanatory fragments [6]. Hammer's analysis of student epistemologies in physics showed that students' beliefs about the nature of physical knowledge, whether equations are definitions, laws, or calculational rules, significantly affect how they learn [7]. The present framework provides a systematic vocabulary for the distinctions that both teachers and students are already navigating, often without naming them.

Case study: $\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0$

This equation illustrates why the two axes cannot be collapsed.

If $\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}$, then

$$\boldsymbol{\nabla} \cdot \boldsymbol{B} = \boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \boldsymbol{A}) = 0$$

by a vector-calculus identity. The vanishing divergence is then a mathematical consequence of the potential representation; its epistemic status is *identity*.

Yet the physical availability of a single, globally defined vector potential rests on the empirical absence of magnetic monopoles. If magnetic charges existed, the source equation would take the form

$$\boldsymbol{\nabla} \cdot \boldsymbol{B} = \rho_m^{\text{mag}},$$

and the global vector-potential representation would fail. In this reading, $\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0$ is an empirical law asserting the non-existence of magnetic charge, and its epistemic status is *empirical law*.

The architectural role remains the same in both readings: it is a *structural constraint* within the field theory. What changes is the epistemic status, which depends on whether the equation is read as a consequence of a representation or as a physical assertion that the representation is available in the first place.

This is not a marginal subtlety. Dirac's 1931 theory of magnetic monopoles showed that quantisation of electric charge could follow from the existence of even a single monopole [10]. The equation $\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0$ thereby encodes a physical claim about the charge spectrum of the universe, dressed in the notation of a vector identity.

Case study: $\boldsymbol{F} = m\boldsymbol{a}$

Newton's second law is the most familiar equation in physics and among the most epistemically ambiguous. In different contexts it has been read as:

- A **definition of force**: force is whatever produces $m\boldsymbol{a}$. In this reading, the equation is an identity that introduces a quantity. Ernst Mach advocated a version of this position.
- A **definition of inertial mass**: mass is the ratio F/a for a body subject to a known force. This is the operational reading.
- An **empirical law**: given independent definitions of force, mass and acceleration, the equation asserts an experimentally discovered proportionality.
- A **constitutive plus dynamical fusion**: Paper I showed that $\boldsymbol{F} = m\boldsymbol{a}$ compresses three distinct modules: the kinematic definition $\boldsymbol{v} = \dot{\boldsymbol{r}}$, the constitutive map $\boldsymbol{p} = m\boldsymbol{v}$, and the dynamical balance $\dot{\boldsymbol{p}} = \boldsymbol{F}$ [1].

Newton himself was not entirely consistent on the matter. The *Principia* defines "quantity of motion" as the product of mass and velocity (a constitutive-definitional act) and then states the second law as a relation between impressed force and the change of motion (a dynamical act). The compression into $\boldsymbol{F} = m\boldsymbol{a}$ is a later convenience that obscures the original two-step structure.

The classroom consequence is direct. Redish observes that students often read equations as calculational procedures rather than as physical assertions [8]. A student who treats $\boldsymbol{F} = m\boldsymbol{a}$ as a single, self-interpreting rule has no reason to ask whether the equation is defining force, defining mass, or asserting an empirical relationship. The two-axis classification makes the question visible and answerable, context by context.

Case study: $pV = nRT$

The ideal gas law occupies the constitutive position in thermodynamics, connecting macroscopic state variables through a material-response relation. Its epistemic status, however, shifts depending on the context:

- As an **empirical law**: it summarises the experimental behaviour of dilute gases, combining Boyle's, Charles's and Avogadro's laws into a single relation. Here it is a constitutive empirical law.
- As an **idealisation**: it describes a gas of non-interacting point particles. Real gases deviate; the van der Waals equation $\left(p + a/V^2\right)(V - b) = nRT$ restores two neglected effects. Here $pV = nRT$ is a constitutive idealisation.
- As a **definition of temperature**: in statistical mechanics, temperature may be defined through $1/T = \partial S/\partial E$, and the ideal gas law then becomes a derived result for a system of non-interacting particles. Its status shifts from input to output.
- As a **constitutive closure in the hinge framework**: Paper I noted that Hooke's law, $F = -kx$, is literally constitutive, describing how a material system responds to deformation. The equation of state $pV = nRT$ plays the same structural role for a thermal system: it is the phenomenological information that conservation laws alone do not supply [1].

In every case, the architectural role is constitutive. What changes is the epistemic status, from empirical law to idealisation to derived result, depending on which other statements are taken as primitive.

Case study: the second law of thermodynamics

The second law illustrates how epistemic status can be genuinely contested rather than merely context-dependent.

- In Clausius's formulation, heat does not spontaneously flow from a colder to a hotter body. This is an empirical generalisation.
- In Kelvin's formulation, no cyclic process extracts heat from a single reservoir and converts it entirely to work. This is a constraint on possible processes.
- In the entropy formulation, $\mathrm{d}S \geq \delta Q/T$ for any process, with equality for reversible processes. Here it is a balance inequality.
- In statistical mechanics, the second law emerges as an overwhelmingly probable consequence of the microscopic dynamics of large systems. Here it is a derived statistical result.
- In some axiomatic treatments, entropy is defined by the second law, making it a definition.

The architectural role is consistent across formulations: the second law functions as a dynamical or balance constraint governing the direction of processes. The epistemic status ranges from empirical law to statistical theorem to definitional axiom. A student who asks "is the second law a law, a definition, or an approximation?" is asking a well-formed question that the single-axis classification cannot answer. The two-axis matrix locates the question precisely: the architectural coordinate is fixed while the epistemic coordinate varies with the formulation.

Equations that change status

The case studies above are not isolated curiosities. Equations routinely shift epistemic status under changes of formulation or theoretical context. A systematic account would recognise several mechanisms:

1. **Representation adoption.** An equation that is an empirical law becomes an identity after a representation is adopted that encodes it. Example: $\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0$ after introduction of $\boldsymbol{A}$.
2. **Theory change.** A constitutive law in one theory becomes a limiting case in its successor. Example: $\boldsymbol{p} = m\boldsymbol{v}$ is a constitutive law in classical mechanics and an approximation in relativistic mechanics.
3. **Change of primitive.** An equation that defines one quantity in terms of others may become a derived result when a different quantity is taken as primitive. Example: $pV = nRT$ is an empirical law when T is measured independently, but a derived result in statistical mechanics where T is defined through $\partial S/\partial E$.
4. **Scope change.** A universal law may become an idealisation when its domain of validity is restricted. Example: $pV = nRT$ is a law for ideal gases and an idealisation for real gases at high pressure.
5. **Pedagogical framing.** The same equation may be presented as a definition in one course and as an empirical law in another, depending on which quantities have been established first. Hammer's studies of student epistemologies showed that such framing choices significantly affect how students reason with the equation [7].

A disciplined interrogation

The two-axis classification refines the hinge idea developed in Papers I and II [1, 4]. A hinge relation preserves one module of the theoretical grammar while leaving another slot open. The present classification adds that the hinge relation and its completion may occupy different epistemic positions.

Consider $v = f\lambda$. Architecturally, it is a structure equation: it relates three kinematic quantities describing a periodic travelling wave. Epistemically, it is a definition or kinematic identity, true by virtue of what v, f and λ mean. The completion, a medium law such as $v = \sqrt{T/\mu}$, occupies the constitutive position architecturally and the empirical-law position epistemically: it encodes physical content about the string that the kinematic identity does not contain.

The hinge works as a pedagogical device precisely because the two slots have different epistemic authorities. The invariant part is definitional or structural, providing an organising framework that does not depend on the specific physical case. The case-specific part is empirical or constitutive, providing the information that makes the model physically determinate. Students who can identify this asymmetry are doing what Sherin calls activating a symbolic form [9]: they recognise that the equation has an internal structure in which different parts carry different kinds of authority.

The two axes together suggest a compact set of questions for reading any physical equation:

1. **Architectural role.** Where does this equation sit in the theory? Is it a structure equation, a constitutive equation, a dynamical or balance equation, an interaction equation, or a derived composite?
2. **Epistemic status.** What kind of claim is this equation making? Is it a definition, an identity, an empirical law, a constitutive model, a conservation law, an approximation, an idealisation, a boundary condition, or a derived result?
3. **Stability under reformulation.** Would a change of representation, a change of primitive, or a change of theory alter the architectural role, the epistemic status, or both?
4. **Open slots.** Does the equation leave a theoretical slot unfilled? If so, what kind of completion is required: constitutive, interactional, geometric, conditional, or systemic?
5. **Compression.** Does the equation fuse several modules? If so, which architectural positions and which epistemic statuses have been compressed into a single line?

These questions will form the mathematical derivation. An equation becomes visible as a component with a determinate architectural position and a context-dependent epistemic authority. The equals sign no longer conceals the difference between a definition, a law, a model and a consequence.

Conclusion

Tonti's grammar classifies equations by their architectural role within a physical theory: where they sit and which variable families they connect. The present essay adds a second, orthogonal classification: the epistemic status of the equation, what kind of claim it makes about the world.

The two axes are independent. Equations sharing an architectural role may have different epistemic statuses. Equations sharing an epistemic status may occupy different architectural positions. The most instructive cases are equations that shift position on one axis or both under a change of formulation, a change of theory, or a change of pedagogical context.

Paper I therefore had logical priority: an equation occupies a position within a theory before we ask what authority it carries there. The present paper adds that second coordinate. Architectural role answers where the equation sits; epistemic status answers what kind of warrant, scope and revisability it possesses.

The matrix has direct consequences for physics teaching. The hinge framework of Paper II [4] works because the invariant module and the case-specific completion occupy different epistemic positions: the first is typically definitional or structural, the second typically empirical or constitutive. Making this asymmetry explicit gives students a vocabulary for asking not only "what equation do I need?" but "what kind of equation is this, and what kind of authority does it carry?"

A formula sheet presents equations as if they were all the same kind of thing. They are not. Definitions, identities, empirical laws, constitutive assumptions, interaction couplings, boundary conditions and idealisations differ in origin, in authority, in domain of validity, and in what happens when they fail. The two-axis classification makes those differences visible.

References

[1] L. McCulloch-James, "A grammar of physical theories: structure, constitution and interaction in physical models," Paper I of III, 2025. Available: leelemacjames.github.io.

[2] E. Tonti, "On the formal structure of physical theories," *Quaderni dei Gruppi di Ricerca Matematica del CNR*, Istituto di Matematica del Politecnico di Milano, 1975.

[3] E. Tonti, *The Mathematical Structure of Classical and Relativistic Physics: A General Classification Diagram*, Modeling and Simulation in Science, Engineering and Technology, New York: Birkhäuser/Springer, 2013. doi: 10.1007/978-1-4614-7422-7.

[4] L. McCulloch-James, "Teaching through the hinge," Paper II of III, 2025. Available: leelemacjames.github.io.

[5] L. McCulloch-James, "The placeholder force: How the same F conceals different physics," 2025. Available: leelemacjames.github.io.

[6] A. A. diSessa, "Toward an epistemology of physics," *Cognition and Instruction*, vol. 10, nos. 2–3, pp. 105–225, 1993. doi: 10.1207/s1532690xci1002&3_2.

[7] D. Hammer, "Epistemological beliefs in introductory physics," *Cognition and Instruction*, vol. 12, no. 2, pp. 151–183, 1994. doi: 10.1207/s1532690xci1202_4.

[8] E. F. Redish, "Problem solving and the use of math in physics courses," in *Proceedings of the Conference World View on Physics Education in 2005: Focusing on Change*, Delhi, India, pp. 1–10, 2005. Available: arXiv:physics/0608268.

[9] B. L. Sherin, "How students understand physics equations," *Cognition and Instruction*, vol. 19, no. 4, pp. 479–541, 2001. doi: 10.1207/S1532690XCI1904_3.

[10] P. A. M. Dirac, "Quantised singularities in the electromagnetic field," *Proceedings of the Royal Society A*, vol. 133, no. 821, pp. 60–72, 1931. doi: 10.1098/rspa.1931.0130.